QRAEMON

Product–Market Fit Analysis

Early-Stage Product Strategy, Customer Problem Validation
and Beachhead Market Definition

The Persistent Experience Layer for Physical Intelligence

Converting robot experience into better future decisions

Strategic Research Document

Version 1.0
October 2026

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1 Executive Summary

Core conclusion:
QRAEMON has the characteristics of a potentially important deep-technology company, but its current broad vision does not constitute product–market fit by itself.
The strongest early commercial opportunity is not to sell "an Artificial Brain" or "robot memory." The strongest beachhead is a narrowly defined system that enables deployed robots to use previous successful and failed experiences to improve future decisions, particularly in variable manipulation tasks where failures require human intervention or engineering effort.

1.1 Current PMF Assessment

Dimension	Assessment	Interpretation
Problem existence	9/10	Repeated robot failures and adaptation costs are real problems.
Market growth	9/10	Robotics and Physical AI are expanding rapidly.
Timing	9/10	Foundation models, sensors, compute and robotics are converging.
Technical feasibility	7/10	A bounded cognitive layer is technically plausible.
Differentiation	6/10	"Memory" alone is insufficient differentiation.
Defensibility	7/10	Potentially strong if experience becomes decision-linked.
Willingness to pay	6/10	Strong if measurable economic value is demonstrated.
Integration difficulty	5/10	Real-world robotics integration remains significant.
Distribution	5/10	Requires direct technical sales and design partners.
Current PMF	4/10	Must still be empirically established.
Long-term opportunity	10/10	Potentially very large if the core mechanism works.

1.2 Strategic Verdict

Recommended first product:

QRAEMON Experience & Recovery Engine

The product should initially solve a narrow problem:

Repeated physical failures that require human intervention or robot reprogramming.

The Artificial Brain remains the long-term destination. The experience engine is the commercial and scientific entry point.

2 Product Definition

2.1 Long-Term Vision

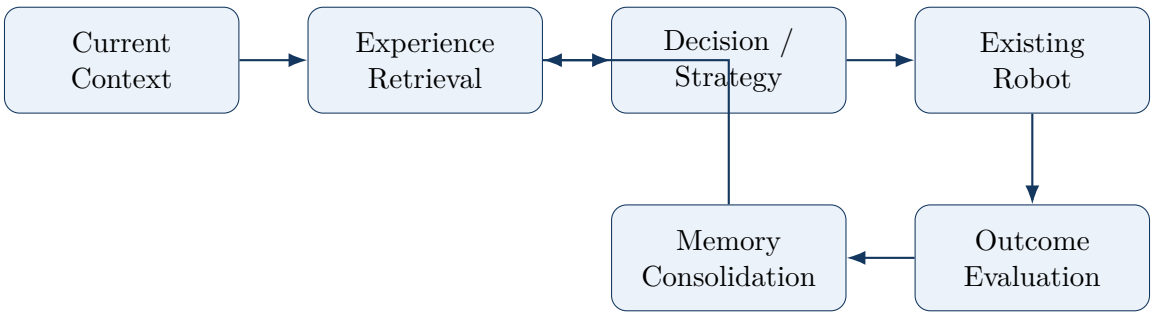
QRAEMON is intended to become a lifelong, continually learning artificial cognitive architecture that can preserve identity, accumulate experience, reorganize memory, reason, adapt and transfer knowledge across physical embodiments.

The long-term thesis is:

One evolving cognitive system rather than a sequence of independently retrained models.

2.2 Initial Product

The first product should deliberately be much narrower.



2.3 Product Promise

The initial product promise should be:

QRAEMON helps deployed robots learn from their operational experience so that the same type of failure does not have to be solved repeatedly.

This is substantially more commercially concrete than claiming general robot intelligence.

3 Customer Problem

3.1 The Operational Problem

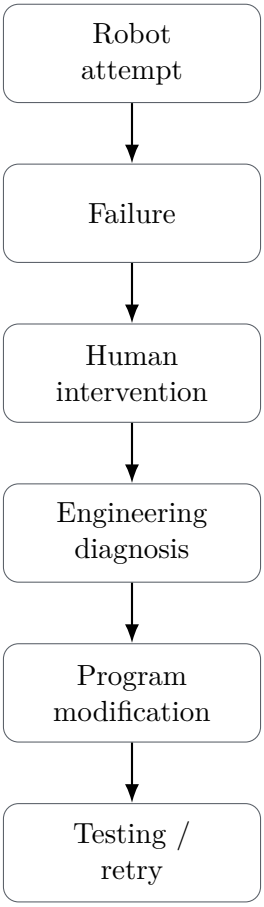
Robots are highly effective when their environment and task distribution are sufficiently constrained. The economic problem increases when:

- object conditions vary;
- task conditions change;
- unexpected failures occur;
- the robot encounters previously unseen situations;
- humans must intervene;
- engineers must diagnose failures;
- robot programs must be modified;
- new testing and commissioning are required.

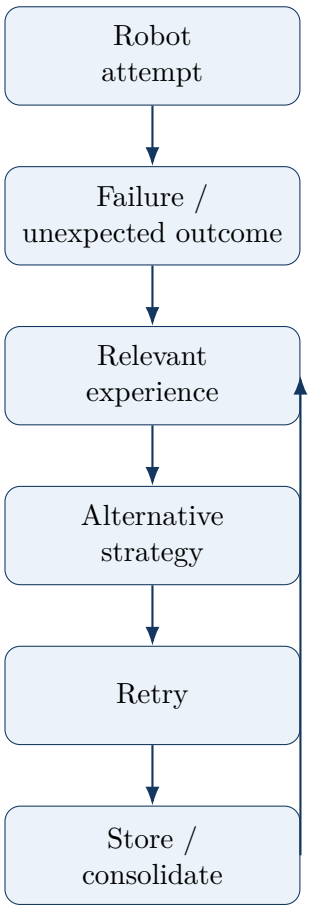
The commercial pain is therefore not "lack of memory."
It is:

Repeated operational cost caused by the robot failing to exploit previous experience.

3.2 Current Workflow



QRAEMON seeks to transform this into:



4 Beachhead Market

4.1 Recommended Beachhead

Primary beachhead:

Robotics companies operating variable manipulation tasks

These customers should already possess:

- operational robotic hardware;
- an existing control stack;
- repeated physical tasks;
- measurable failure or intervention events;
- engineering resources;
- willingness to modify the software stack.

4.2 Why This Beachhead

This segment is attractive because it provides:

1. immediate access to physical robots;

2. a clear baseline;
3. measurable outcomes;
4. repeated experiences;
5. high-value failure events;
6. technical sophistication;
7. potential for rapid experimentation;
8. a direct path toward broader robotics deployment.

4.3 Beachhead Scoring

Segment	Pain	Budget	Integration	Pilot Speed	Strategic	Assessment
Robotics startups	5	4	4	5	5	**Best initial target**
Industrial R&D	5	5	3	3	5	Strong second target
Advanced manufacturing	5	5	2	2	5	Strong but slower
Robot integrators	4	4	2	4	4	Distribution opportunity
Research laboratories	4	2	5	5	4	Excellent design partners
Humanoid companies	5	5	1	1	5	Long-term opportunity

5 Jobs-to-be-Done

The initial customer is effectively hiring QRAEMON to perform five jobs:

1. **Reduce repeated failures.**
2. **Reduce human intervention.**
3. **Reduce engineering reprogramming.**
4. **Adapt to recurring environmental variation.**
5. **Convert operational experience into future performance.**

The customer is **not** primarily hiring QRAEMON to obtain a new AI model.

6 10× Hypothesis

6.1 Technical Hypothesis

QRAEMON hypothesis:
A robot equipped with selective, decision-linked experience can improve task adaptation, recovery and transfer without requiring complete retraining of the underlying robot policy or controller.

6.2 Commercial Hypothesis

The commercially relevant hypothesis is:

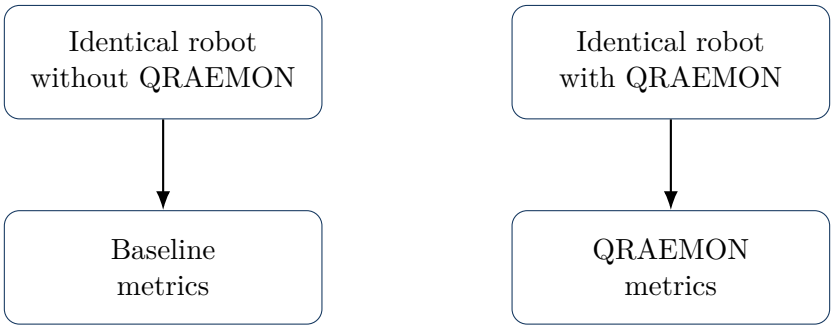
Less intervention + less reprogramming + less downtime

The initial product must therefore demonstrate a measurable economic improvement.

7 Product–Market Fit Experiment

7.1 Controlled Comparison

The most important early experiment is:



7.2 Primary Metrics

Metric	Purpose
Task success rate	Measures final task performance.
Recovery success rate	Measures ability to recover from failures.
Human intervention rate	Measures operational burden.
Trials to adaptation	Measures learning efficiency.
Task completion time	Measures operational performance.
Engineering intervention hours	Measures deployment cost.
Retrieval precision	Measures memory relevance.
Transfer performance	Measures generalization of experience.
Memory growth	Measures long-term scalability.
Stability / interference	Measures continual-learning safety.

7.3 Core Scientific Metric

A useful central metric is:

$$DIE = P(\text{success} \mid \text{experience}) - P(\text{success} \mid \text{no experience})$$

where DIE represents *Decision Improvement from Experience*.
The objective is not merely to demonstrate that QRAEMON stores experiences.
The objective is to demonstrate that:

$$\boxed{\text{Experience} \rightarrow \text{Better Decision} \rightarrow \text{Better Physical Outcome}}$$

8 Willingness to Pay

The first commercial model should be based on measurable value rather than compute consumption.

8.1 Initial Pricing Hypothesis

Stage	Indicative Pricing Hypothesis	Purpose
Research pilot	\$10k–\$30k	Technical validation
Industrial pilot	\$25k–\$75k	Economic validation
Production deployment	\$5k–\$25k+/workcell/year	Recurring revenue
Enterprise deployment	\$100k+/year	Fleet / enterprise licensing

These figures are initial hypotheses for customer discovery and should not be treated as validated market prices.

8.2 Value-Based Pricing Logic

A simplified ROI model is:

$$ROI = \frac{\text{avoided intervention cost} + \text{avoided downtime} + \text{avoided engineering cost}}{\text{QRAEMON cost}}$$

QRAEMON should become easier to purchase when the customer can quantify this equation.

9 Customer Discovery

The first 20–30 interviews should target:

- robotics CTOs;
- robotics software engineers;
- robotics deployment engineers;
- automation engineers;
- manufacturing R&D managers;
- system integrators;
- robotics laboratory researchers.

The interviews should focus on:

1. What failures occur repeatedly?
2. How often do they occur?
3. Who resolves them?
4. How long does resolution take?
5. What information is retained after resolution?
6. Does the robot benefit from the previous solution?

7. What does each failure cost?
8. Would a system that reduced intervention be purchased?

Avoid asking:

"Would you use a robot with memory?"

Instead ask:

"What happens when this robot encounters a failure it has seen before?"

10 Competitive Risk

The central competitive threat is not another "robot memory startup."

The larger threat is that foundation-model and robotics-platform companies incorporate continual experience directly into their systems.

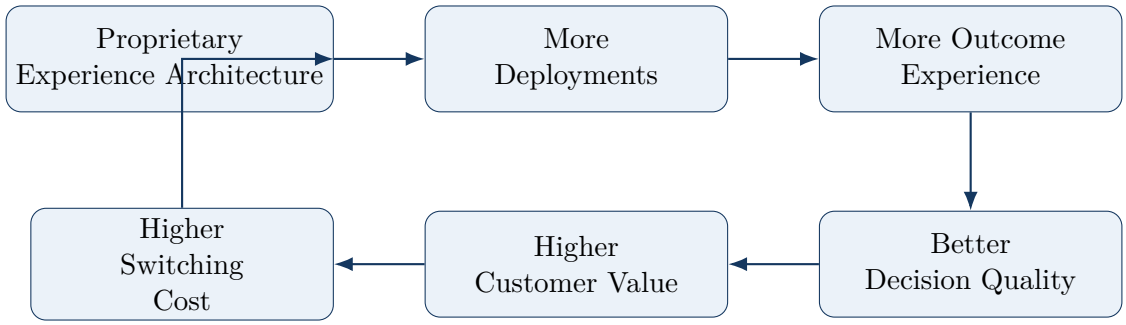
Therefore QRAEMON cannot depend on memory storage alone.

Its defensibility must emerge from:

- proprietary experience representation;
- decision-linked retrieval;
- outcome evaluation;
- selective consolidation;
- forgetting;
- cross-task transfer;
- cross-robot transfer;
- deployment data;
- evaluation infrastructure;
- embodiment-independent architecture.

11 Defensibility

The intended moat should compound as deployments increase.



12 PMF Failure Risks

The following assumptions must be falsified rather than assumed:

1. Memory may not materially improve decisions.
2. Retrieval may select irrelevant experiences.
3. Experience may fail to transfer between tasks.
4. Continual learning may create instability.
5. Customers may prefer simpler policy updates.
6. Integration cost may exceed economic value.
7. Robot manufacturers may incorporate the capability internally.
8. Foundation models may make the architecture unnecessary.

13 PMF-Oriented Product Roadmap

Stage	Question	Success Condition
0–3 months	Can experience change decisions?	Measurable decision improvement
3–6 months	Can decisions improve physical outcomes?	Better task performance
6–9 months	Does experience transfer?	Improvement on related new conditions
9–12 months	Will a customer pay?	Paid pilot
12–18 months	Can deployment be repeated?	Repeatable integration package
18–24 months	Can the system scale?	Multiple robots / tasks

14 Final Product–Market Fit Verdict

Current conclusion:
QRAEMON should not enter the market as a general "Artificial Brain."
It should enter through a narrow, measurable problem:

Reducing repeated robot failures and human intervention through persistent operational experience.

The first product should therefore be an Experience and Recovery Engine for variable robotic manipulation.

14.1 Strategic Sequence

Experience Engine → Failure Recovery → Continual Skill Adaptation → Cross-Task Transfer → Cross-Robot E

The Artificial Brain is the destination.
The experience engine is the beachhead.